

MENU-DRIVEN PROGRAM USING LOOPS AND SWITCH STATEMENT

Multiplication, Addition, Subtraction and Division Tables

Name	Rudra Pratap Singh
SAP ID	590032811
Batch	16

1. Aim

To develop a menu-driven C program that accepts two positive integers m and n ($m \leq n$) and uses nested loops with a switch statement to print multiplication, addition, subtraction and division tables for every integer from m to n .

2. Problem Statement

Accept two positive integers m and n such that $m \leq n$. Repeatedly display a five-option menu. According to the user's choice, print the requested table for each integer from m to n . The program must continue until option 5 (Exit) is selected and must show an error message for an invalid menu choice.

3. Concepts Used

- Input/output using `printf()` and `scanf()`
- Input validation using while loops
- Menu repetition using do-while loop
- Decision making using switch-case
- Nested for loops for values from m to n and 1 to 10
- Type casting to float and `%.2f` formatting for division results

4. Algorithm

1. Start.
2. Declare integer variables m , n , choice, i and j .
3. Read m and n .
4. If m or n is non-positive, display an error and read the values again.
5. If $m > n$, display an error and read the values again.
6. Display the menu with choices 1 to 5 and read choice.
7. Use a switch statement on choice.
8. Case 1: For each i from m to n , use j from 1 to 10 and print $i \times j$.
9. Case 2: For each i from m to n , use j from 1 to 10 and print $i + j$.
10. Case 3: For each i from m to n , use j from 1 to 10 and print $i - j$.
11. Case 4: For each i from m to n , use j from 1 to 10 and print i / j as a floating-point value rounded to two decimal places.
12. Case 5: Display an exit message.
13. Default: Display "Invalid menu choice".
14. Repeat the menu until choice is 5.
15. Stop.

5. Program Logic / Flow

Step	Flow
1	Start → Input m and n
2	Validate: $m > 0$, $n > 0$ and $m \leq n$
3	Display menu → Read choice
4	switch(choice) → execute selected table
5	Choice 1–4 → return to menu
6	Invalid choice → error → return to menu
7	Choice 5 → Exit → Stop

6. Test-Case Table

TC	Input / Choice	Type	Expected Result
1	m=2, n=3; choice=1	Valid	Print first 10 multiples of 2 and 3.
2	m=4, n=4; choice=2	Boundary	Print addition table only for 4, from +1 to +10.
3	m=1, n=2; choice=3	Valid	Print subtraction tables for 1 and 2.
4	m=5, n=6; choice=4	Valid	Print division tables with values to 2 decimal places.
5	m=3, n=3; choice=5	Boundary	Exit without performing a table operation.
6	m=0, n=5	Invalid	Reject input because both integers must be positive.
7	m=-2, n=4	Invalid	Reject input because m is not positive.
8	m=8, n=3	Invalid	Reject input because m > n.
9	m=2, n=3; choice=9	Invalid	Display invalid menu choice and show menu again.
10	m=1, n=1; choices=1,2,3,4,5	Boundary	All four operations work for the smallest positive equal range; then exit.

7. C Program

```
#include <stdio.h>

int main()
{
    int m, n, choice, i, j;

    /* Input and validation of m and n */
    while (1)
    {
        printf("Enter two positive integers m and n (m <= n): ");
        scanf("%d %d", &m, &n);

        if (m <= 0 || n <= 0)
            printf("Error: m and n must be positive integers.\n\n");
        else if (m > n)
            printf("Error: m must be less than or equal to n.\n\n");
        else
            break;
    }

    do
    {
        printf("\n===== MENU =====\n");
        printf("1. Multiplication tables\n");
        printf("2. Addition tables\n");
        printf("3. Subtraction tables\n");
        printf("4. Division tables\n");
        printf("5. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice)
        {
            case 1:
                for (i = m; i <= n; i++)
                {
                    printf("\nMultiplication table of %d:\n", i);
                    for (j = 1; j <= 10; j++)
                        printf("%d x %d = %d\n", i, j, i * j);
                }
                break;

            case 2:
```

```

        for (i = m; i <= n; i++)
        {
            printf("\nAddition table of %d:\n", i);
            for (j = 1; j <= 10; j++)
                printf("%d + %d = %d\n", i, j, i + j);
        }
        break;

    case 3:
        for (i = m; i <= n; i++)
        {
            printf("\nSubtraction table of %d:\n", i);
            for (j = 1; j <= 10; j++)
                printf("%d - %d = %d\n", i, j, i - j);
        }
        break;

    case 4:
        for (i = m; i <= n; i++)
        {
            printf("\nDivision table of %d:\n", i);
            for (j = 1; j <= 10; j++)
                printf("%d / %d = %.2f\n", i, j,
                    (float)i / j);
        }
        break;

    case 5:
        printf("Exiting program. Thank you!\n");
        break;

    default:
        printf("Error: Invalid menu choice. Please choose 1 to 5.\n");
    }
} while (choice != 5);

return 0;
}

```

8. Compilation and Execution

The program can be compiled using a standard C compiler. Example command:

```
gcc menu_tables.c -o menu_tables
```

Then execute the program and enter the test data. A representative execution is shown below.

9. Sample Execution

Enter two positive integers m and n (m <= n): 2 3

```
===== MENU =====
```

1. Multiplication tables
2. Addition tables
3. Subtraction tables
4. Division tables
5. Exit

Enter your choice: 4

Division table of 2:

```
2 / 1 = 2.00
2 / 2 = 1.00
2 / 3 = 0.67
...
2 / 10 = 0.20
```

Division table of 3:

```
3 / 1 = 3.00
3 / 2 = 1.50
3 / 3 = 1.00
...
3 / 10 = 0.30
```

```
===== MENU =====
```

```
...
```

Enter your choice: 9

Error: Invalid menu choice. Please choose 1 to 5.

```
===== MENU =====
```

```
...
```

Enter your choice: 5

Exiting program. Thank you!

10. Explanation

The outer for loop selects each integer from m to n, while the inner for loop runs from 1 to 10. The switch statement selects the required operation. A do-while loop keeps the menu active until the user chooses 5. In the division case, (float)i / j forces floating-point division and %.2f displays exactly two digits after the decimal point.

11. Result

The menu-driven C program was prepared to perform multiplication, addition, subtraction and division table operations for all integers from m to n. It validates the input range, handles invalid menu choices, repeats the menu correctly, and terminates only when the Exit option is selected.